

An AI-assisted note on the dp-rank of ordered abelian groups

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Introduction

This note presents a proof of the known computation of the dp-rank of ordered abelian groups in the pure language of ordered groups. Gurevich and Schmitt showed that every ordered abelian group has NIP [5], and hence has dp-rank at most $\aleph_0$. For background on NIP and dp-rank, we refer to Simon's guide [8].

We recall the relevant definitions. Let T be a complete theory in a language $\mathcal{L}$, and work in a monster model. An *ICT-pattern* of depth κ consists of formulas $(\varphi_\alpha(x, \bar{y}_\alpha))_{\alpha < \kappa}$, parameters $(\bar{b}_{\alpha,k})_{\alpha < \kappa, k < \omega}$, and elements (a_η) indexed by functions $\eta : \kappa \rightarrow \omega$, such that

$$\models \varphi_\alpha(a_\eta, \bar{b}_{\alpha,k}) \iff \eta(\alpha) = k$$

for all $\alpha < \kappa$, all $k < \omega$, and all such η . The *dp-rank* of T is at least κ if an ICT-pattern of depth κ exists. NIP is equivalent to dp-rank at most $|\mathcal{L}| + \aleph_0$; since L_{oag} is countable, this recovers the bound $\aleph_0$ already cited.

Let G be an ordered abelian group. If a spine $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank of G is $\aleph_0$. If instead G is nontrivial and has finite spines, then

$$\text{dp-rk}(G) = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Here $\mathbb{P}$ is the set of primes, and $\mathcal{S}_p^\infty$ is the set of nonempty $H \in \mathcal{S}_p$ for which the rib $R_H^{\mathfrak{s}_p}$ is infinite. None of these results is new: the formula was obtained independently by Farré [4] and by Halevi and Hasson [6]. Dolich and Goodrick [2] independently characterized the strongly dependent ordered abelian groups, that is, those of dp-rank less than $\aleph_0$.

The purpose of this note is to practice AI-assisted proof writing. The author supplied sketches of the proofs; ChatGPT 5.6 Sol and Grok 4.6 expanded them into complete proofs, which the author subsequently reviewed and revised. The author takes full responsibility for the contents. We leave the basics of the model theory of ordered abelian groups, and the notation used below, to [7, Section 2], which is based on Cluckers and Halupczok [1].

1 The infinite spine case

Throughout this note, G denotes an ordered abelian group. For $n \geq 2$ and $a \notin nG$, recall that $\mathfrak{s}_n(a)$ is the largest convex subgroup $H \in G$ such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$, and we regard $\emptyset$ as being smaller than every convex subgroup. We set $\mathcal{S}_n = \{\mathfrak{s}_n(a) \mid a \in G\}$. The subgroups $\mathfrak{s}_n(a)$ are uniformly definable in a [1, Lemma 2.1]. Consequently, comparisons and equalities between values of the maps $\mathfrak{s}_n$ are formulas in L_{oag} .

Lemma 1.1 ([7, Lemma 2.8(2) and Remark 2.6]). *Let p be a prime, $r \geq 1$, and $a, b \in G$.*

1. *If $0 \leq j < r$, then $\mathfrak{s}_{p^r}(p^j a) = \mathfrak{s}_{p^{r-j}}(a)$. If $j \geq r$, then $\mathfrak{s}_{p^r}(p^j a) = \emptyset$.*
2. *If $\mathfrak{s}_{p^r}(a) \neq \mathfrak{s}_{p^r}(b)$, then $\mathfrak{s}_{p^r}(a + b) = \max\{\mathfrak{s}_{p^r}(a), \mathfrak{s}_{p^r}(b)\}$.*

Lemma 1.2. *Let p be a prime and let $H < H'$ be nonempty elements of $\mathcal{S}_p$. There exists $a \in G$ such that $\mathfrak{s}_p(a) = H$ and $\mathfrak{s}_{p^r}(a) < H'$ for every $r \geq 1$.*

Proof. Choose $b \in G$ such that $\mathfrak{s}_p(b) = H$. Since $H' > H$, the maximality of H gives $b \in H' + pG$. We may therefore write $b = a + pc$ with $a \in H'$ and $c \in G$. As $a - b \in pG$, we have $\mathfrak{s}_p(a) = \mathfrak{s}_p(b) = H$. If $\mathfrak{s}_{p^r}(a) \geq H'$ for some $r \geq 1$, then $a \in H' \subseteq \mathfrak{s}_{p^r}(a)$, contrary to $a \notin \mathfrak{s}_{p^r}(a) + p^r G$. Therefore $\mathfrak{s}_{p^r}(a) < H'$. ■

Lemma 1.3 (Finite coding). *Let p be a prime and $K_0 < K_1 < \dots < K_N$ be nonempty elements of $\mathcal{S}_p$. There exists $x \in G$ such that $\mathfrak{s}_{p^{i+1}}(x) = K_i$ for every $0 \leq i \leq N$.*

Proof. For each $j < N$, use Lemma 1.2 to choose $a_j \in G$ such that $\mathfrak{s}_p(a_j) = K_j$ and $\mathfrak{s}_{p^r}(a_j) < K_{j+1}$ for every $r \geq 1$. Also choose $a_N \in G$ with $\mathfrak{s}_p(a_N) = K_N$, and set

$$x = a_0 + pa_1 + \dots + p^N a_N.$$

Fix $0 \leq i \leq N$. By Lemma 1.1, $\mathfrak{s}_{p^{i+1}}(p^i a_i) = \mathfrak{s}_p(a_i) = K_i$. If $j < i$, then $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \mathfrak{s}_{p^{i+1-j}}(a_j) < K_{j+1} \leq K_i$, while, if $j > i$, then $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \emptyset$. Thus, among the summands of x , the unique maximal $\mathfrak{s}_{p^{i+1}}$ -value is K_i . Repeatedly applying Lemma 1.1(2) gives $\mathfrak{s}_{p^{i+1}}(x) = K_i$. ■

Theorem 1.4. *If $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank of G is $\aleph_0$.*

Proof. We work in a sufficiently saturated monster model of $\text{Th}(G)$. Since $\mathcal{S}_p$ is an infinite interpretable linear order, compactness allows us to choose elements $(b_{i,k})_{i,k < \omega}$ such that all the values $H_{i,k} := \mathfrak{s}_p(b_{i,k})$ are nonempty and

$$H_{i,k} < H_{j,l} \quad \text{whenever} \quad i < j \text{ or } (i = j \text{ and } k < l).$$

For a function $\eta : \omega \rightarrow \omega$, consider the partial type

$$\Sigma_\eta(x) = \{ \mathfrak{s}_{p^{i+1}}(x) = H_{i,\eta(i)} \mid i < \omega \}.$$

This type is finitely satisfiable. Indeed, given $N < \omega$, the values $H_{0,\eta(0)} < H_{1,\eta(1)} < \dots < H_{N,\eta(N)}$ form a finite increasing chain, so Lemma 1.3 realizes the first $N + 1$ conditions of Σ_η . By saturation, there is $a_\eta \in G$ realizing Σ_η .

For every $i < \omega$, let $\varphi_i(x; y)$ be the formula $\mathfrak{s}_{p^{i+1}}(x) = \mathfrak{s}_p(y)$. As noted above, this is an L_{oag} -formula. For all $\eta : \omega \rightarrow \omega$, $i < \omega$, and $k < \omega$, we have

$$\models \varphi_i(a_\eta; b_{i,k}) \iff H_{i,\eta(i)} = H_{i,k} \iff \eta(i) = k.$$

Hence, these formulas and parameters form an ICT-pattern of depth $\aleph_0$, and therefore the dp-rank of G is at least $\aleph_0$.

Every ordered abelian group has NIP [5]. Since L_{oag} is countable, this gives dp-rank at most $\aleph_0$. Consequently, the dp-rank of G is exactly $\aleph_0$. $\blacksquare$

2 The lower bound in the finite-spine case

We now assume that G is nontrivial and has finite spines: $\mathcal{S}_p$ is finite for every prime p . For a prime p and $H \in \mathcal{S}_p \setminus \{\emptyset\}$, recall that

$$H^{[p]} = \bigcap_{K \in G, K > H} (K + pG) \quad \text{and} \quad R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

by [7, Definition 2.10 and Lemma 2.14(1)]. We also use $\mathcal{S}_{p^r} = \mathcal{S}_p$ for every $r \geq 1$, as established in [1, Lemma 2.2(1)]. We put

$$\mathcal{S}_p^\infty = \{ H \in \mathcal{S}_p \setminus \{\emptyset\} \mid |R_H^{\mathfrak{s}_p}| = \infty \}.$$

Lemma 2.1. *Let p be a prime and $H \in \mathcal{S}_p^\infty$. There is a sequence $(a^k)_{k < \omega}$ such that $\mathfrak{s}_p(a^k) = \mathfrak{s}_p(a^k - a^l) = H$ whenever $k \neq l$. If $H < K$ for some $K \in \mathcal{S}_p \setminus \{\emptyset\}$, the sequence may moreover be chosen inside K .*

Proof. Choose elements $(u^k)_{k < \omega}$ whose images in $R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$ are nonzero and pairwise distinct. Thus, neither u^k nor $u^k - u^l$ belongs to $H + pG$, while all these elements belong to $H^{[p]}$. By [7, Lemma 2.14(1)], this says precisely that $\mathfrak{s}_p(u^k) = \mathfrak{s}_p(u^k - u^l) = H$ for $k \neq l$.

Suppose that $H < K$. Since $u^k \in H^{[p]} \subseteq K + pG$, we can write $u^k = a^k + pc_k$ with $a^k \in K$. Replacing u^k by a^k does not change its image in $R_H^{\mathfrak{s}_p}$. The images therefore remain nonzero and pairwise distinct, so the required equalities remain true. $\blacksquare$

Lemma 2.2 (Simultaneous coding for one prime). *Let p be a prime and let $H_0 < H_1 < \dots < H_{n-1}$ be elements of $\mathcal{S}_p^\infty$. For each $i < n$, there is a sequence $(a_i^k)_{k < \omega}$ as in Lemma 2.1 for H_i , and these sequences can be chosen so that $a_i^k \in H_{i+1}$ whenever $i < n-1$. For $\zeta : n \rightarrow \omega$, set*

$$b_\zeta = a_0^{\zeta(0)} + pa_1^{\zeta(1)} + \dots + p^{n-1}a_{n-1}^{\zeta(n-1)}.$$

Then, for every $i < n$ and every $k < \omega$,

$$b_\zeta - p^i a_i^k \in H_i + p^{i+1}G \iff k = \zeta(i).$$

Proof. The required sequences are supplied by Lemma 2.1. For $j < i$, we have $a_j^{\zeta(j)} \in H_{j+1} \subseteq H_i$, whereas for $j > i$, the term $p^j a_j^{\zeta(j)}$ belongs to $p^{i+1}G$. Hence $b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G$.

If $k \neq \zeta(i)$, then $\mathfrak{s}_{p^{i+1}}(p^i(a_i^{\zeta(i)} - a_i^k)) = \mathfrak{s}_p(a_i^{\zeta(i)} - a_i^k) = H_i$, so $p^i a_i^{\zeta(i)} - p^i a_i^k \notin H_i + p^{i+1}G$. Together with the preceding congruence, this proves the claim. ■

The following standard consequences of compactness will be used without proof. For the second assertion, finite satisfiability follows from the Chinese remainder theorem [7, Lemma 2.1(2)].

Lemma 2.3. *Let G be a sufficiently saturated monster model of the theory of a nontrivial ordered abelian group.*

1. *The subgroup $\bigcap_{m \geq 2} mG$ is cofinal and coinitial in G .*
2. *Let $P_0 \subseteq \mathbb{P}$ and $(a_p)_{p \in P_0}$ be a sequence from G . There exists $a \in G$ such that $a - a_p \in p^r G$ for every $p \in P_0$ and $r \geq 1$.*

Theorem 2.4. *Let G be a nontrivial ordered abelian group with finite spines. Then*

$$\text{dp-rk}(G) \geq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Proof. We work in a sufficiently saturated monster model. For each prime p , put $n_p = |\mathcal{S}_p^\infty|$ and, when $n_p > 0$, enumerate $\mathcal{S}_p^\infty = \{H_{p,0} < \dots < H_{p,n_p-1}\}$. The number n_p is finite because G has finite spines. Let $\mathcal{I} = \{(p, i) \mid p \in \mathbb{P}, 0 \leq i < n_p\}$. Thus $|\mathcal{I}| = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$, and $\mathcal{I}$ is finite or countable.

For each p with $n_p > 0$, choose sequences $(a_{p,i}^k)_{k < \omega}$ as in Lemma 2.2, and denote the corresponding coding elements by $b_{p,\zeta}$ for $\zeta : n_p \rightarrow \omega$. For $(p, i) \in \mathcal{I}$, let $\varphi_{p,i}(x; y)$ be the formula $x - y \in \mathfrak{s}_{p^{i+1}}(y) + p^{i+1}G$. This is an L_{oag} -formula by the uniform definability of $\mathfrak{s}_{p^{i+1}}$. Lemma 1.1 implies $\mathfrak{s}_{p^{i+1}}(p^i a_{p,i}^k) = \mathfrak{s}_p(a_{p,i}^k) = H_{p,i}$ for every $k < \omega$, and hence $\models \varphi_{p,i}(b_{p,\zeta}; p^i a_{p,i}^k) \iff k = \zeta(i)$.

Fix a function $\eta : \mathcal{I} \rightarrow \omega$. For each p with $n_p > 0$, let $\eta_p : n_p \rightarrow \omega$ be given by $\eta_p(i) = \eta(p, i)$. By Lemma 2.3(2), there is $b_\eta \in G$ such that $b_\eta - b_{p,\eta_p} \in p^{i+1}G$ for every p with $n_p > 0$ and $i \geq 0$. Since translation by an element of $p^{i+1}G$ does not change the truth of formula $\varphi_{p,i}$, we have $\models \varphi_{p,i}(b_\eta; p^i a_{p,i}^k) \iff k = \eta(p, i)$ for all $(p, i) \in \mathcal{I}$ and $k < \omega$.

It remains to add the row arising from the order. Since the monster is sufficiently saturated, there exists $c > 0$ such that $-c < b_\eta < c$ for every $\eta : \mathcal{I} \rightarrow \omega$. By Lemma 2.3(1), choose $t_0 < t_1 < \dots$ from $\bigcap_{m \geq 2} mG$ so that $t_{k+1} - t_k > 2c$ for every $k < \omega$. Set $I_k = (t_k - c, t_k + c)$. These intervals are pairwise disjoint and satisfy $I_0 < I_1 < \dots$. We take $\psi(x; y, z) := (y < x < z)$ for the additional row, with parameter sequence $(t_k - c, t_k + c)_{k < \omega}$.

Let $\nu : \{*\} \cup \mathcal{I} \rightarrow \omega$, and put $\eta = \nu|_{\mathcal{I}}$. The element $a_\nu = b_\eta + t_{\nu(*)}$ belongs to $I_{\nu(*)}$. Moreover, $t_{\nu(*)} \in p^{i+1}G$ for every $(p, i) \in \mathcal{I}$, so translation by $t_{\nu(*)}$ does not change the truth of any formula $\varphi_{p,i}$. It follows that

$$\begin{aligned} \models \psi(a_\nu; t_k - c, t_k + c) &\iff \nu(*) = k, \\ \models \varphi_{p,i}(a_\nu; p^i a_{p,i}^k) &\iff \nu(p, i) = k. \end{aligned}$$

We have therefore constructed an ICT-pattern of depth

$$1 + |\mathcal{I}| = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

as required. ■

3 The upper bound in the finite-spine case

We continue to assume that G is nontrivial and has finite spines, and now also assume that

$$N = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. We use the following standard criterion.

Fact 3.1 (Indiscernible-sequence criterion). *Let T be a complete theory, and let $n \geq 1$. Fix a Dedekind complete $I \models DLO$. Suppose that, whenever $(\bar{a}_i)_{i \in I}$ is an indiscernible sequence and b is a singleton, one can divide I by at most n points so that*

$$\text{tp}(\bar{a}_i/b) = \text{tp}(\bar{a}_j/b)$$

whenever i and j lie in the same convex component of the remainder. Then $\text{dp-rk}(T) \leq n$.

This may be proved in the same way as [3, Fact 2.10]. Henceforth I denotes this Dedekind complete index set. For $p \in \mathbb{P}$ and $a \in G$, recall that $\mathfrak{t}_p(a) = \bigcup_{H \in \mathcal{S}_p, a \notin H} H$. We set $\mathcal{T}_p = \{\mathfrak{t}_p(a) \mid a \in G\}$, but this coincides with $\mathcal{S}_p$ if G has finite spines. We will also use the following result originating in the relative quantifier elimination by Cluckers and Halupczok [1]. See [7, Definition 2.20] for the notation.

Proposition 3.2. *Let G be an ordered abelian group with finite spines and let f be a partial map from G to G whose domain is a subgroup. Suppose that f is*

an ordered group embedding and, for every a in its domain, every prime p , and every $r \geq 1$, the following hold:

$$\begin{aligned} \mathfrak{s}_{p^r}(f(a)) &= \mathfrak{s}_{p^r}(a), & \mathfrak{t}_p(f(a)) &= \mathfrak{t}_p(a), \\ a =_{\bullet} k_{\bullet} &\iff f(a) =_{\bullet} k_{\bullet}, & a \equiv_{p^r, \bullet} k_{\bullet} &\iff f(a) \equiv_{p^r, \bullet} k_{\bullet}. \end{aligned}$$

Here the third condition is required for every $k \in \mathbb{Z} \setminus \{0\}$, and the fourth for every $1 \leq k < p^r$. Then f is elementary.

Proof. This is a special case of [7, Proposition 2.26]. We use the identity map on all the auxiliary sorts in the cited proposition. We explain why predicates $D_{p^r}^{[p^s]}$, where $s \geq r$, require no additional hypothesis. Put $H = \mathfrak{s}_{p^r}(a)$. If $H = \emptyset$ then $D_{p^r}^{[p^s]}(a)$ does not hold. Suppose that $H \neq \emptyset$. If H has a successor H' in $\mathcal{S}_p$ then $H^{[p^s]} = H' + p^s G$, while if H has no successor then $H^{[p^s]} = G$. In either case, it automatically follows that $a \in H^{[p^s]} + p^r G$, which implies $D_{p^r}^{[p^s]}(a)$. This completes the proof. $\blacksquare$

Lemma 3.3 (see the proof of [7, Lemma 3.4]). *Let p be a prime, $H \in \mathcal{S}_p \setminus \{\emptyset\}$, and $r \geq 1$. If $R_H^{\mathfrak{s}_p}$ is finite, then $R_H^{\mathfrak{s}_{p^r}}$ is finite. Consequently,*

$$|R_H^{\mathfrak{s}_{p^r}}| = \infty \implies H \in \mathcal{S}_p^{\infty}.$$

The next lemma isolates the single contribution coming from the order and from the ordered ribs associated with the maps $\mathfrak{t}_p$.

Lemma 3.4 (The common order cut). *Let $(\bar{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. One can divide I by at most one point so that the following data are constant on each convex component of the remainder:*

1. the sign of $\bar{m} \cdot \bar{a}_i - mb$;
2. $\mathfrak{t}_p(\bar{m} \cdot \bar{a}_i - mb)$;
3. the truth value of $\bar{m} \cdot \bar{a}_i - mb =_{\bullet} k_{\bullet}$.

Here the assertions hold simultaneously for all integer tuples $\bar{m}$, all nonzero integers m , all primes p , and all nonzero integers k .

Proof. Let $\mathcal{H}$ consist of $\{0\}$ and all nonempty elements of the finite sorts $\mathcal{T}_p$. Finiteness of every $\mathcal{T}_p$ implies that an indiscernible sequence remains indiscernible with respect to formulas involving elements of $\mathcal{H}$.

Fix $H \in \mathcal{H}$, an integer tuple $\bar{m}$, a nonzero integer m , and an integer k , where either $k = 0$ or G/H is discrete. Let

$$c_i = (\bar{m} \cdot \bar{a}_i)/H - k_H \in G/H.$$

$(c_i)_{i \in I}$ is also indiscernible. In particular, it is either constant or strictly monotone. Consequently, exactly one of the following holds:

- (i) the sign of $c_i - m(b/H)$ is constant as i varies in I ;

- (ii) there exists $i_0 \in I$ such that the terms $c_i - m(b/H)$ for $i < i_0$ are all negative and those terms for $i > i_0$ are all positive, or vice versa.

(The realization of the cut as a point of I uses Dedekind completeness.) Now suppose that (ii) holds with cutting point i_0 , and let $K \in \mathcal{H}$ satisfy $K > H$. Then $k_H/K = 0$ and hence $c_i/K = (\overline{m} \cdot \overline{a}_i)/K$. Thus, either (i) or (ii) holds also for $c_i/K - m(b/K)$. If (ii) holds, then the cutting point is the same i_0 because $c_i/K - m(b/K)$ for $i < i_0$ are nonpositive and those for $i > i_0$ are nonnegative, or vice versa.

We next show that two distinct points of I cannot arise as cutting points in (ii). Let $c_i = \overline{m} \cdot \overline{a}_i/H - k_H$ and $d_i = \overline{m}' \cdot \overline{a}_i/K - k'_K$ with $H \leq K$. Suppose that distinct i_0 and j_0 in I arise as cutting points of $c_i - m(b/H)$ and $d_i - m'(b/K)$ in (ii), respectively. We may assume that $i_0 < j_0$. We also assume that $c_i - m(b/H)$ for $i < i_0$ are negative by negating the sequence if necessary. Furthermore, we may assume that $m = m'$ by multiplying both sequences by some nonzero integers if necessary. Choose

$$h < i_0 < r < s < j_0 < l.$$

If (i) holds for $c_i/K - m(b/K)$, then $c_i/K = m(b/K)$ for all i . Hence the signs of $d_s - c_{i_0}/K$ and $d_l - c_{i_0}/K$ are different, contradicting the indiscernibility of $(c_i/K, d_i)_{i \in I}$. Therefore we assume that (ii) holds for $c_i/K - m(b/K)$ with cutting point i_0 , and denote c_i/K by c_i henceforth. If (d_i) is also increasing, then we have $c_r > d_s$ but $c_h < d_l$, contradicting indiscernibility. If instead (d_i) is decreasing, then $c_h < d_h$ but $c_l > d_l$, again leading to a contradiction.

Thus at most one cutting point of I arises. Dividing I by this point if it exists, the signs under consideration are constant on each convex component of the remainder. Taking $H = \{0\}$ and $k = 0$ gives (1). (2) follows because $\mathfrak{t}_p(x) < H$ if and only if $x \in H$ if and only if $x/H = 0$ for $H \in \mathcal{T}_p$. Finally, by [7, Lemma 2.21(1)], a witness to $x = \bullet k_\bullet$ is $\mathfrak{t}_2(x)$. Thus (2), together with constancy of the sign of $x/H - k_H$, gives (3). ■

We now control all contributions from the $\mathfrak{s}_{p^r}$ -side. Let $(\overline{a}_i)_{i \in I}$ and b be as in Lemma 3.4. For a prime p and for $H \in \mathcal{S}_p \setminus \{\emptyset\}$, call $i_0 \in I$ a (p, H) -exception if there are an integer tuple $\overline{m}$, a nonzero integer m , an integer $r \geq 1$, and an integer k such that either $k = 0$ or G/H is discrete, and such that, writing

$$c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H \in G/H,$$

one has

$$c_i \equiv_{p^r} m(b/H) \iff i = i_0.$$

Lemma 3.5 (Uniqueness of the congruence exception). *Let p be a prime and let $H \in \mathcal{S}_p \setminus \{\emptyset\}$.*

1. *There is at most one (p, H) -exception.*
2. *If i_0 is a (p, H) -exception, then i_0 is also a (p, K) -exception for some $K \in \mathcal{S}_p^\infty$.*

Proof. For (1), suppose that $i_0 < j_0$ are (p, H) -exceptions, witnessed respectively by

$$(c_i, m, r) \quad \text{and} \quad (d_i, m', r')$$

in the notation above. Convex subgroups are pure, so G/H is torsion-free, and thus in G/H one has $x \equiv_{p^s} 0 \iff p^t x \equiv_{p^{s+t}} 0$ for all $s, t \geq 1$. This allows us to assume that $r = r'$. One also has $x \equiv_{p^s} 0 \iff ux \equiv_{p^s} 0$ whenever u is coprime to p . By writing $m = np^e$ where n is coprime to p and multiplying the inverse of n in $\mathbb{Z}/p^r\mathbb{Z}$, we also may assume that $m = p^e$, and similarly that $m' = p^f$. We now have

$$\begin{aligned} c_i \equiv_{p^r} p^e(b/H) &\iff i = i_0, \\ d_i \equiv_{p^r} p^f(b/H) &\iff i = j_0. \end{aligned}$$

If $e \leq f$, eliminating b gives

$$d_{j_0} - p^{f-e} c_{i_0} \equiv_{p^r} 0.$$

By indiscernibility, the same relation holds with j_0 replaced by any $j > i_0$. Subtracting yields $d_j \equiv_{p^r} d_{j_0}$, so the second exceptional congruence also holds at j , a contradiction. The case $e > f$ is symmetric.

For (2), suppose again (c_i, m, r) witnesses that i_0 is a (p, H) -exception, where $c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H$. By indiscernibility, the value of $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_j - \overline{m} \cdot \overline{a}_i)$ is constant when distinct i, j range over I . Let K be this value. We have $K \geq H$ because otherwise, $c_i \equiv_{p^r} c_j$ for all distinct i, j , which contradicts the definition of i_0 . It follows that $c_{i_0}/K \equiv_{p^r} m(b/K)$. $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - \overline{m} \cdot \overline{a}_{i_0}) = K$ implies $c_i/K \not\equiv_{p^r} c_{i_0}/K$ for $i \neq i_0$, and thus $c_i/K \not\equiv_{p^r} m(b/K)$, which shows that i_0 is also a (p, K) -exception. The images of $(\overline{m} \cdot \overline{a}_i - \overline{m} \cdot \overline{a}_{i_0})_{i \neq i_0}$ in $R_K^{\mathfrak{s}_{p^r}}$ are pairwise distinct, hence $K \in \mathcal{S}_p^\infty$ by Lemma 3.3. $\blacksquare$

Lemma 3.6 (Control of the $\mathfrak{s}$ -data). *For every pair (p, H) with $H \in \mathcal{S}_p^\infty$, choose its exceptional index if one exists, and let E be the set of indices so obtained. Then $|E| \leq N$. If $C \subseteq I$ is convex and $C \cap E = \emptyset$, then for any $i, j \in C$, every integer tuple $\overline{m}$, every nonzero integer m , every prime p , every $r \geq 1$, and every $1 \leq k < p^r$, we have*

$$\begin{aligned} \mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - mb) &= \mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_j - mb), \\ \overline{m} \cdot \overline{a}_i - mb \equiv_{p^r, \bullet} k_\bullet &\iff \overline{m} \cdot \overline{a}_j - mb \equiv_{p^r, \bullet} k_\bullet. \end{aligned}$$

Proof. By Lemma 3.5(1) one has $|E| \leq N$. By part (2) of the same lemma, every (p, H) -exception already arises as a (p, K) -exception for some $K \in \mathcal{S}_p^\infty$, hence belongs to E .

Fix a prime p , a nonempty $H \in \mathcal{S}_p$, an integer tuple $\overline{m}$, a nonzero integer m , an integer $r \geq 1$, and an integer k such that either $k = 0$ or G/H is discrete. Put

$$c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H \in G/H.$$

We first show that

$$c_i \equiv_{p^r} m(b/H) \iff c_j \equiv_{p^r} m(b/H)$$

for all $i, j \in C$. By indiscernibility, either $c_i \equiv_{p^r} c_j$ for all distinct $i, j \in I$, or else $c_i \not\equiv_{p^r} c_j$ for all distinct $i, j \in I$. The claim is trivial in the first case. In the second case, there is at most one $i_0 \in I$ such that $c_{i_0} \equiv_{p^r} m(b/H)$. If this i_0 exists, then i_0 is a (p, H) -exception, so $i_0 \in E$ and the claim follows.

Applying this with $k = 0$ as H varies, we obtain that $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - mb)$ is constant on C . If the common value is $\emptyset$, the predicate $\overline{m} \cdot \overline{a}_i - mb \equiv_{p^r, \bullet} k_\bullet$ fails on C . Otherwise, by [7, Lemma 2.21(2)], a witness to the predicate is this common value H . If G/H is dense, the predicate fails on C ; if G/H is discrete, the equivalence above proves the constancy of the predicate on C . ■

Theorem 3.7. *Let G be a nontrivial ordered abelian group with finite spines, and suppose that*

$$N = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. Then

$$\text{dp-rk}(G) \leq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Proof. We work in a sufficiently saturated monster model. Let $(\overline{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. Apply Lemma 3.4 and Lemma 3.6. Dividing I by the common order-cut point together with the at most N points of E produces at most $N + 1$ points.

Fix a convex component C of the remainder, and let $i, j \in C$. Let M_i be the subgroup generated by the entries of $\overline{a}_i$ together with b , and define

$$f_{ij} : M_i \longrightarrow M_j$$

by

$$f_{ij}(\overline{m} \cdot \overline{a}_i + mb) = \overline{m} \cdot \overline{a}_j + mb.$$

The case $m = 0$ in the following verifications follows directly from indiscernibility. For $m \neq 0$, Lemma 3.4(1) shows that the sign of every element is preserved. Note that $\overline{m} \cdot \overline{a}_i + mb = \overline{m}' \cdot \overline{a}_i + m'b$ if and only if $(\overline{m} - \overline{m}') \cdot \overline{a}_i + (m - m')b = 0$. This shows that f_{ij} is well-defined and injective. It also follows that f_{ij} is an ordered group embedding. The same lemma, parts (2) and (3), shows that f_{ij} preserves every $\mathfrak{t}_p$ and every predicate $x =_\bullet k_\bullet$. By Lemma 3.6, it preserves every $\mathfrak{s}_{p^r}$ and every predicate $x \equiv_{p^r, \bullet} k_\bullet$. Therefore, Proposition 3.2 implies that f_{ij} is elementary. Since it fixes b and sends $\overline{a}_i$ to $\overline{a}_j$, we obtain

$$\text{tp}(\overline{a}_i/b) = \text{tp}(\overline{a}_j/b).$$

We have verified the hypothesis of Fact 3.1 with $n = N + 1$. Hence $\text{dp-rk}(G) \leq N + 1$, as required. ■

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